進階程式設計
Advanced Program Design

L2

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資料型態(C89)

Data Type	Keyword	qualifier	Final definition	Memory (Bytes)	Range
Character	char		char	1	-128 to 127
		unsigned	unsigned char	1	0 to 255
Integer	int		int	2	-32,768 to 32,767
		unsigned	unsigned int	2	0 to 65535
		signed	signed int	2	-32,768 to 32,767
		short	short int	2	-32,768 to 32,767
		unsigned short	unsigned short int	2	0 to 65535
		signed short	signed short int	2	-32,768 to 32,767
		long	long int	4	-2,147,483,648 to 2,147,483,647
		unsigned long	unsigned long int	4	0 to 4,294,967,295
		signed long	signed long int	4	-2,147,483,648 to 2,147,483,647
Decimal	float		float	4	3.4E-38 to 3.4E+38
	double		double	8	1.7E-308 to 1.7E+308
		long	long double	10	3.4E-4932 to 1.1E+4932
	Integer(C99)	unsigned long long	unsigned long long int	8	
		long long	long long int	8	

變數Variables、常數Constant

- 有號變數(signed)與無號變數(unsigned)的差異
- 變數生命週期的差異
  - 區域變數(Local variables)
  - 全域變數(Global variables) //須節制使用，不易除錯
  - 靜態變數(Static variables)
- 變數應給予初始值(不能假設內容會是0)
  - int variable_A; // int variable_A = 0;
  - char myT; // char myT = ‘\0’;
- 常數會佔記憶體空間(好處是編譯期間會提示常數被更動)，用#define取代

American Standard Code for Information Interchange, ASCII
美國標準資訊交換碼

Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII
0	00000000	000	00	NUL	32	00100000	040	20	SP	64	01000000	100	40	@	96	01100000	140	60	`
1	00000001	001	01	SOH	33	00100001	041	21	!	65	01000001	101	41	A	97	01100001	141	61	a
2	00000010	002	02	STX	34	00100010	042	22	"	66	01000010	102	42	B	98	01100010	142	62	b
3	00000011	003	03	ETX	35	00100011	043	23	#	67	01000011	103	43	C	99	01100011	143	63	c
4	00000100	004	04	EOT	36	00100100	044	24	$	68	01000100	104	44	D	100	01100100	144	64	d
5	00000101	005	05	ENQ	37	00100101	045	25	%	69	01000101	105	45	E	101	01100101	145	65	e
6	00000110	006	06	ACK	38	00100110	046	26	&	70	01000110	106	46	F	102	01100110	146	66	f
7	00000111	007	07	BEL	39	00100111	047	27	'	71	01000111	107	47	G	103	01100111	147	67	g
8	00001000	010	08	BS	40	00101000	050	28	(	72	01001000	110	48	H	104	01101000	150	68	h
9	00001001	011	09	HT	41	00101001	051	29	)	73	01001001	111	49	I	105	01101001	151	69	i
10	00001010	012	0A	LF	42	00101010	052	2A	*	74	01001010	112	4A	J	106	01101010	152	6A	j
11	00001011	013	0B	VT	43	00101011	053	2B	+	75	01001011	113	4B	K	107	01101011	153	6B	k
12	00001100	014	0C	FF	44	00101100	054	2C	,	76	01001100	114	4C	L	108	01101100	154	6C	l
13	00001101	015	0D	CR	45	00101101	055	2D	-	77	01001101	115	4D	M	109	01101101	155	6D	m
14	00001110	016	0E	SO	46	00101110	056	2E	.	78	01001110	116	4E	N	110	01101110	156	6E	n
15	00001111	017	0F	SI	47	00101111	057	2F	/	79	01001111	117	4F	O	111	01101111	157	6F	o
16	00010000	020	10	DLE	48	00110000	060	30	0	80	01010000	120	50	P	112	01110000	160	70	p
17	00010001	021	11	DC1	49	00110001	061	31	1	81	01010001	121	51	Q	113	01110001	161	71	q
18	00010010	022	12	DC2	50	00110010	062	32	2	82	01010010	122	52	R	114	01110010	162	72	r
19	00010011	023	13	DC3	51	00110011	063	33	3	83	01010011	123	53	S	115	01110011	163	73	s
20	00010100	024	14	DC4	52	00110100	064	34	4	84	01010100	124	54	T	116	01110100	164	74	t
21	00010101	025	15	NAK	53	00110101	065	35	5	85	01010101	125	55	U	117	01110101	165	75	u
22	00010110	026	16	SYN	54	00110110	066	36	6	86	01010110	126	56	V	118	01110110	166	76	v
23	00010111	027	17	ETB	55	00110111	067	37	7	87	01010111	127	57	W	119	01110111	167	77	w
24	00011000	030	18	CAN	56	00111000	070	38	8	88	01011000	130	58	X	120	01111000	170	78	x
25	00011001	031	19	EM	57	00111001	071	39	9	89	01011001	131	59	Y	121	01111001	171	79	y
26	00011010	032	1A	SUB	58	00111010	072	3A	:	90	01011010	132	5A	Z	122	01111010	172	7A	z
27	00011011	033	1B	ESC	59	00111011	073	3B	;	91	01011011	133	5B	[	123	01111011	173	7B	{
28	00011100	034	1C	FS	60	00111100	074	3C	<	92	01011100	134	5C	\	124	01111100	174	7C	|
29	00011101	035	1D	GS	61	00111101	075	3D	=	93	01011101	135	5D	]	125	01111101	175	7D	}
30	00011110	036	1E	RS	62	00111110	076	3E	>	94	01011110	136	5E	^	126	01111110	176	7E	~
31	00011111	037	1F	US	63	00111111	077	3F	?	95	01011111	137	5F	_	127	01111111	177	7F	DEL

判斷式/迴圈

- 非零為真、零為假
  - 真：if(1)、if(10)、if(-2)、if(!0)、if(~0)
  - 假：if(0)

//==========

- if…
- If…else
- if…else if…
- switch…case

//==========

- while()
- do…while()
- for()

//==========

- break/goto/continue

位元處理運算元

Symbol	Operator
&	bitwise AND
|	bitwise OR
^	bitwise XOR (exclusive OR)
<<	left shift
>>	right shift
~	bitwise NOT

```
a = 5, b = 9
a & b = 1
a | b = 13
a ^ b = 12
~a = -6
b << 1 = 18
b >> 1 = 4
```

```c
#include <stdio.h>
int main() {
    // a = 5(00000101), b = 9(00001001)
    unsigned char a = 5, b = 9;

    // The result is 00000001
    printf("a = %d, b = %d\n", a, b);
    printf("a&b = %d\n", a & b);

    // The result is 00001101
    printf("a|b = %d\n", a | b);

    // The result is 00001100
    printf("a^b = %d\n", a ^ b);

    // The result is 11111010
    printf("~a = %d\n", a = ~a);

    // The result is 00010010
    printf("b<<1 = %d\n", b << 1);

    // The result is 00000100
    printf("b>>1 = %d\n", b >> 1);

    return 0;

}
```

函式

- Prototype(原型宣告)的意義
  - void func(void);
  - int func(void); // char func(void); // float func(void); …
  - void func(int, int); // void func(char, int); //void func(float, char); ….
  - int func(int, int);
- void func(int, int, int, int, int, int); //參數列3個參數即可
  - void funcA(int, int); + int funcB(int, int); + int funcC(int, int);
- 差異性
  - void func(int, int);
  - void func(int*, int*);

陣列array

- 差異性1：
  - int arr[10] = {0}; //全部元素初始值為0
  - int arr[10] = {1,2,3,0}; //arr[3]~arr[9]均為0
- 差異性2
  - char arr[] = {‘H’, ’e’, ’l’, ’l’, ‘o’};
  - char arr[] = “Hello”;
  - char arr[5] = “Hello”; //有風險
  - char arr[6] = “Hello”;
- 陣列、索引值、存取超過陣列的範圍是Bug
- 陣列名稱arr ＝ arr起始位址 ＝ arr陣列第0個元素的位址 ＝ &arr[0]

陣列Array

- 2維陣列、3維陣列
  - int arr[2][3] = {{1,2,3},{7,8,9}};
  - int arr[3][3][3];
- 記憶體是一維架構

Computer		Programmers		
Address	Content	Name	Type	Value
90000000	00	sum	int (4 bytes)	000000FF(255_{10})
90000001	00			
90000002	00			
90000003	FF			
90000004	FF	age	short (2 bytes)	FFFF(-1_{10})
90000005	FF			
90000006	1F	averge	double (8 bytes)	1FFFFFFFFFFFFFFF ($4.45015E-308_{10}$)
90000007	FF			
90000008	FF			
90000009	FF			
9000000A	FF			
9000000B	FF			
9000000C	FF			
9000000D	FF			
9000000E	90	ptrSum	int* (4 bytes)	90000000
9000000F	00			
90000010	00			
90000011	00			

Note: All numbers in hexadecimal

判斷式/迴圈

- 試讀取月份(1~12)，然後判斷季節(3~5月春季、6~8月夏季、9~11月秋季、12~2月為冬季)
- 試讀取5個學生的成績，成績在0~59分為C，60~75分為B，76~100分為A，最後將得到A、B、C的人數印出。

//==========

- 找出最小的k值，使得1+2+3+…+k≤m。
- 判斷輸入的整數是否為質數？
- 輸入整數，求其所有的因數。(例如24，則印出1,2,3,4,6,8,12和24)

The End